PROJECT DOCUMENTATION

Brain Tumor Distribution Analysis

An interactive Power BI dashboard on patient demographics, tumor types and affected brain regions

Tool	Microsoft Power BI Desktop
Source file	project.pbix
Report canvas	Single page, 1920 × 1080 px (Fit to page), dark grayscale theme
Fact table	Brain Tumor Data
Supporting table	Tumor Images
Custom visuals	Chiclet Slicer, Image visual (cloudScope)

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1. Project Overview

1.1 Purpose

This project analyses the distribution of brain tumor cases across patient age groups, tumor types, affected brain regions and years of diagnosis. The dashboard gives a quick, single-screen view of who is affected, where tumors occur and how case volumes change over time, with MRI imagery and tumor-grade icons to keep the clinical context visible.

1.2 Objectives

- Summarise the patient population through three KPI cards: total patients, average age and average tumor size.
- Show how cases split between tumor types (Glioma and Medulloblastoma).
- Identify the most affected brain regions and the most affected age groups.
- Track the number of patients diagnosed each year to reveal trends.
- Let users filter by diagnosis status (confirmed or unconfirmed) and by tumor grade.

1.3 Components Used

Component	Details
Data tables	Brain Tumor Data (patient records) and Tumor Images (grade icons and MRI image URLs).
Calculated column	Age Group, which bins patient age into 10-year bands.
DAX measures	Avg Intake py, vs Avg Intake py, Most Affected Group and MABR.
Standard visuals	Cards, line charts, donut chart, clustered bar chart, clustered column chart, text boxes.
Custom / special visuals	Chiclet Slicer (grade selector with icons), Image visual (MRI scan), button-style slicer (diagnosis status).
Design	Custom page background image, Fluent 2 base theme, white-on-charcoal grayscale styling.

2. Dashboard Preview

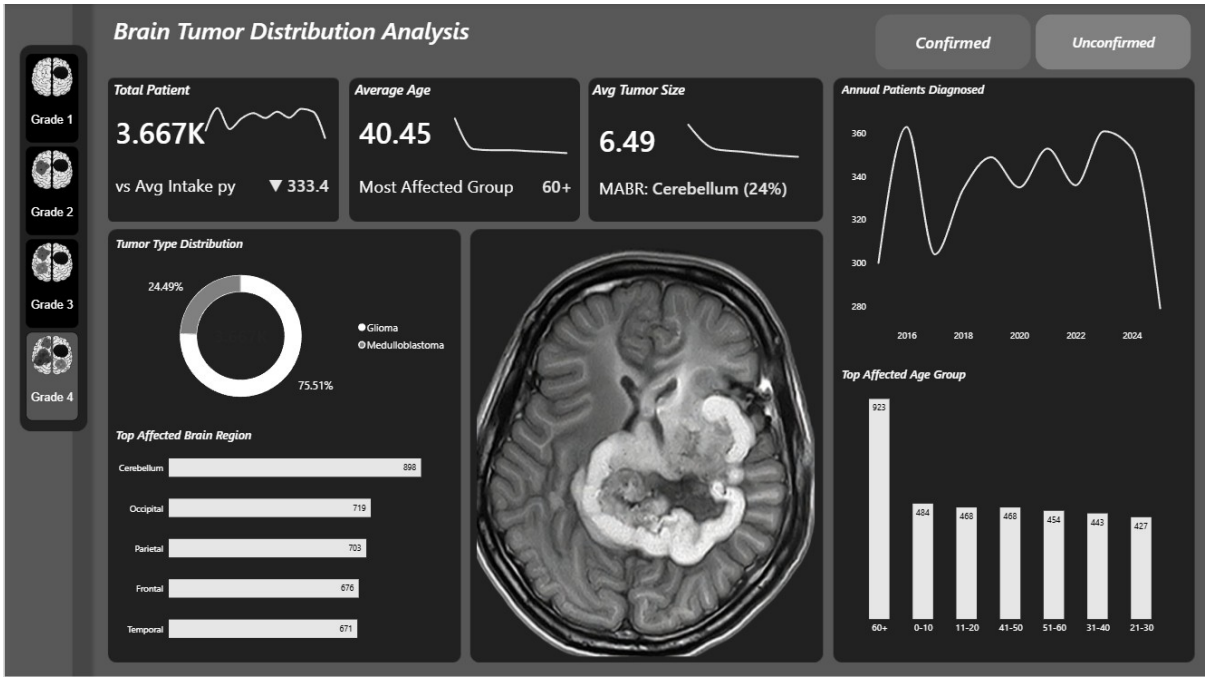


Figure 1. Brain Tumor Distribution Analysis dashboard

The page follows a left-to-right reading flow. Filters sit at the edges (grade slicer on the left, diagnosis toggle at the top right), the KPI cards form the top row, and the supporting breakdowns sit below them around the central MRI image.

Zone	Component	Visual type
Left sidebar	Tumor Grade slicer (Grade 1 to 4)	Chiclet Slicer
Top right	Confirmed / Unconfirmed toggle	Button slicer
Top row (1st card)	Total Patient	Card + line chart + reference-label card
Top row (2nd card)	Average Age	Card + line chart + reference-label card
Top row (3rd card)	Avg Tumor Size	Card + line chart + reference-label card
Middle left	Tumor Type Distribution	Donut chart
Bottom left	Top Affected Brain Region	Clustered bar chart
Centre	MRI scan image	Image visual (cloudScope)
Right, upper	Annual Patients Diagnosed	Line chart
Right, lower	Top Affected Age Group	Clustered column chart

3. Data Model

3.1 Brain Tumor Data (fact table)

Field	Kind	Used for
Patient ID	Column	Distinct count in every patient-count visual and KPI.
Age	Column	Average Age card; source for the Age Group column.
Age Group	Calculated column	Age-group sparkline, Top Affected Age Group chart and Most Affected Group measure.
Date of Diagnosis	Column (year level used)	Total Patient sparkline, Annual Patients Diagnosed chart, Avg Intake py.
Tumor Type	Column	Legend of the donut chart.
Tumor Size (cm)	Column	Average Tumor Size card and its sparkline (sum by region).
Affected Brain Region	Column	Top Affected Brain Region chart, tumor-size sparkline, MABR measure.
Histology Confirmation	Column	Confirmed / Unconfirmed button slicer.

3.2 Tumor Images (supporting table)

Field	Kind	Used for
Tumor Grade	Column	Category of the Chiclet grade slicer.
Icon URL	Column	Brain icon shown on each grade button.
Image URL	Column	MRI scan displayed in the centre of the page.

4. Visual-by-Visual Build Guide

4.1 KPI card pattern

Each of the three KPI cards in the top row is built from three layered visuals so that one tile can show a headline number, a trend and a context label:

- **Main card:** shows the headline metric.
- **Sparkline line chart:** a small line chart with axes and gridlines hidden, placed beside the number to show the trend.
- **Reference-label card (card 2):** a second card using Patient ID (count). In Format visual > Reference label, a label is added and bound to a measure; the card's own value and label are then switched off so only the reference label remains.

4.2 Total Patient (top row, 1st card)

Main card	Patient ID (distinct count).
Sparkline	Line chart: X-axis = Date of Diagnosis (year only), Y-axis = Patient ID (count).
Reference label	Card 2: Patient ID (count) with reference label = measure vs Avg Intake py; value and label disabled.
Shows	Total patients in the current filter context (3.667K), plus the average yearly intake with a trend arrow.

4.3 Average Age (top row, 2nd card)

Main card	Age (average).
Sparkline	Line chart: X-axis = Age Group, Y-axis = Patient ID (count).
Reference label	Card 2: Patient ID (count) with reference label = measure Most Affected Group; value and label disabled.
Shows	Mean patient age (40.45) and the age band with the most patients (60+).

4.4 Average Tumor Size (top row, 3rd card)

Main card	Tumor Size (cm) (average).
Sparkline	Line chart: X-axis = Affected Brain Region, Y-axis = Tumor Size (cm) (sum).
Reference label	Card 2: Patient ID (count) with reference label = measure MABR (Most Affected Brain Region); value and label disabled.
Shows	Mean tumor size (6.49) and the most affected region with its share of cases (Cerebellum, 24%).

4.5 Diagnosis Status Toggle (top right)

Visual	Button slicer.
Field	Histology Confirmation.
Purpose	Splits the page between Confirmed and Unconfirmed patients. Every visual connected to the slicer re-filters on click.

4.6 Tumor Grade Slicer (left sidebar)

Visual	Chiclet Slicer (custom visual).
Fields	Category = Tumor Grade (Tumor Images table); Image = Icon URL (Tumor Images table).
Purpose	Vertical stack of four buttons (Grade 1 to Grade 4), each showing a brain icon, used to select a tumor grade.

4.7 MRI Scan Image (centre)

Visual	Image visual (cloudScope custom visual).
Field	Image URL (Tumor Images table) placed in the Image URL well.
Purpose	Visual centrepiece: displays an MRI scan to anchor the numbers in a clinical image.

4.8 Tumor Type Distribution (middle left)

Visual	Donut chart.
Fields	Legend = Tumor Type; Values = Patient ID (count).
Shows	Proportion of Glioma (75.51%) versus Medulloblastoma (24.49%).

4.9 Top Affected Brain Region (bottom left)

Visual	Clustered bar chart.
Fields	Y-axis = Affected Brain Region; X-axis = Patient ID (count).
Shows	Regions ranked by patient count, highest first (Cerebellum 898).

4.10 Annual Patients Diagnosed (right, upper)

Visual	Line chart.
Fields	X-axis = Date of Diagnosis (year only); Y-axis = Patient ID (count).
Shows	Number of patients diagnosed per year and the shape of the trend over time.

4.11 Top Affected Age Group (right, lower)

Visual	Clustered column chart.
Fields	X-axis = Age Group; Y-axis = Patient ID (count).
Shows	Patient count per age band, with 60+ the largest group (923).

5. DAX Reference

One calculated column and four measures support the dashboard. Each entry below gives the purpose, where it is used, the formula and a short explanation.

5.1 Age Group

Type: Calculated column **Used in:** Age-group sparkline, Top Affected Age Group chart, Most Affected Group measure

```
Age Group =
VAR AgeValue = 'Brain Tumor Data'[Age]
RETURN
SWITCH(
    TRUE(),
    AgeValue >= 0 && AgeValue <= 10, "0-10",
    AgeValue >= 11 && AgeValue <= 20, "11-20",
    AgeValue >= 21 && AgeValue <= 30, "21-30",
    AgeValue >= 31 && AgeValue <= 40, "31-40",
    AgeValue >= 41 && AgeValue <= 50, "41-50",
    AgeValue >= 51 && AgeValue <= 60, "51-60",
    AgeValue > 60, "60+",
    "Unknown"
)
```

- Stores the patient's age in a variable, then SWITCH(TRUE(), ...) returns the label of the first condition that is true.
- Anything that matches no band returns "Unknown", which makes bad or missing ages easy to spot.

5.2 Avg Intake py

Type: Measure **Used in:** Used by vs Avg Intake py

```
Avg Intake py =
VAR _StartYear =
    YEAR ( MIN ( 'Brain Tumor Data'[Date of Diagnosis] ) )
VAR _EndYear =
    YEAR ( MAX ( 'Brain Tumor Data'[Date of Diagnosis] ) )
VAR _NumYears = _EndYear - _StartYear + 1
VAR _TotalPatients =
    CALCULATE ( COUNTROWS ( 'Brain Tumor Data' ) )
RETURN
    DIVIDE ( _TotalPatients, _NumYears )
```

- Finds the first and last diagnosis years in the current filter context and counts the years between them (inclusive).
- Divides total patient rows by that number of years to get the average number of patients admitted per year.
- DIVIDE returns blank instead of an error if the year count is zero.

5.3 vs Avg Intake py

Type: Measure (text) **Used in:** Reference label on the Total Patient card

```
vs Avg Intake py =
VAR _CurrentIntake =
    [Avg Intake py]
VAR _OverallIntake =
    CALCULATE (
        [Avg Intake py],
        ALL ( 'Brain Tumor Data' )
    )
VAR _Diff = _CurrentIntake - _OverallIntake
RETURN
    SWITCH (
        TRUE(),
        _Diff > 0, UNICHAR(9650) & " " & FORMAT(_CurrentIntake, "0.0"),
        _Diff < 0, UNICHAR(9660) & " " & FORMAT(_CurrentIntake, "0.0"),
        FORMAT(_CurrentIntake, "0.0")
    )
```

- Compares the average yearly intake under the current slicer selection with the overall average (ALL removes every filter).
- Returns the current intake as text, prefixed with an up triangle (above overall), a down triangle (below overall) or no symbol (equal).
- The label shows the current yearly average, not the difference itself; the arrow carries the comparison.

5.4 Most Affected Group

Type: Measure (text) **Used in:** Reference label on the Average Age card

```
Most Affected Group =
VAR AgeGroupCounts =
    SUMMARIZE(
        'Brain Tumor Data',
        'Brain Tumor Data'[Age Group],
        "PatientCount", DISTINCTCOUNT('Brain Tumor Data'[Patient ID])
    )
VAR TopGroup =
    TOPN(1, AgeGroupCounts, [PatientCount], DESC)
RETURN
MAXX(TopGroup, 'Brain Tumor Data'[Age Group])
```

- SUMMARIZE builds a small table of age groups with a distinct patient count for each.
- TOPN(1, ..., DESC) keeps the group with the highest count; MAXX extracts its label (here, 60+).

5.5 MABR (Most Affected Brain Region)

Type: Measure (text) **Used in:** Reference label on the Average Tumor Size card

```
MABR =
VAR TotalPatients = COUNTROWS('Brain Tumor Data')
VAR RegionCounts =
    SUMMARIZE(
        'Brain Tumor Data',
        'Brain Tumor Data'[Affected Brain Region],
        "Cnt", COUNTROWS('Brain Tumor Data')
    )
VAR TopRegion = TOPN(1, RegionCounts, [Cnt], DESC)
VAR RegionName = MAXX(TopRegion, 'Brain Tumor Data'[Affected Brain Region])
VAR RegionCount = MAXX(TopRegion, [Cnt])
RETURN
RegionName & " (" & FORMAT(DIVIDE(RegionCount, TotalPatients), "0%") & ")"
```

- Counts rows per brain region, keeps the largest, and divides its count by the total number of patients.
- Returns a single readable string such as "Cerebellum (24%)".

6. Key Findings

The figures below are read from the dashboard as shown in Figure 1.

Metric	Value	Observation
Total patients	3,667	About 333 patients per year on average.
Average age	40.45 years	Cases span all ages, but the 60+ band is the single largest.
Average tumor size	6.49 cm	Cerebellum is the most affected region (24% of patients).
Tumor type	Glioma 75.51% Medulloblastoma 24.49%	Glioma accounts for roughly three of every four cases.
Top regions	Cerebellum 898, Occipital 719, Parietal 703, Frontal 676, Temporal 671	Counts are fairly even across the other four regions; the cerebellum leads by about 180.
Age groups	60+: 923; 0-10: 484; 11-20 and 41-50: 468 each; 51-60: 454; 31-40: 443; 21-30: 427	60+ has almost twice as many patients as any other band; the remaining bands are close to each other.
Annual trend	Roughly 280 to 363 patients per year	Peaks around 2016 and 2023, a dip in 2017, and a sharp fall in the final year shown.

7. How to Use the Dashboard

- **Filter by grade:** click Grade 1 to Grade 4 in the left sidebar.
- **Filter by diagnosis status:** use the Confirmed / Unconfirmed buttons at the top right.
- **Cross-filter:** click a slice of the donut, a bar, a column or a point on the line chart to highlight and filter the other visuals; click again to clear.
- **Read the KPI cards:** the number is the headline metric, the sparkline is its trend, and the small line beneath is the context label.

8. Notes and Possible Improvements

- **Latest-year drop:** the annual chart falls sharply in its last year. If that year is incomplete in the source data, mention it on the page or exclude it so it is not read as a real decline.
- **Grade slicer scope:** Tumor Grade lives in the Tumor Images table. For it to filter the patient visuals, there must be a relationship (or a grade column) linking it to Brain Tumor Data. In Figure 1 the visuals still total 3,667 patients with Grade 4 highlighted, so confirm what the grade selection filters.
- **Age Group bands:** the bands use whole-number boundaries, so an age such as 10.5 would fall into "Unknown". This is not an issue if ages are stored as integers.
- **Measure naming:** vs Avg Intake py returns the current average rather than a difference; renaming it (for example "Intake vs Overall") may avoid confusion.
- **Distinct count vs row count:** patient visuals use a distinct count of Patient ID while Avg Intake py and MABR use COUNTROWS. The results match as long as Patient ID is unique per row.
- **Enhancements:** add a year or date-range slicer, tooltips with tumor size by region, and a second page for tumor-size and grade analysis.